

The University of Nottingham

SCHOOL OF MATHEMATICAL SCIENCES

A LEVEL 3 MODULE, SPRING SEMESTER 2022-2023

TIME SERIES ANALYSIS

Time allowed TWO Hours THIRTY Minutes

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced.

Answer ALL questions

Only a calculator from approved list A may be used in this examination.

List A

Basic Models	Scientific Calculators
Aurora HC133	Aurora AX-582
Casio HS-5D	Casio FX82 family
Deli – DL1654	Casio FX83 family
Sharp EL-233	Casio FX85 family
	Casio FX350 family
	Casio FX570 family
	Casio FX 991 family
	Sharp EL-531 family
	Texas Instruments TI-30 family
	Texas BA II+ family

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 .

(a) For each of the following ARMA processes $\{X_t : t \in \mathbb{Z}\}$: identify the order of the process, write its equation using the backward shift operator, B , and determine whether or not the process is invertible. You should justify your answers and show your working clearly.

(i) $X_t = \frac{1}{2}X_{t-1} - \frac{1}{4}X_{t-2} + Z_t + \frac{1}{3}Z_{t-1}$

(ii) $X_t = \frac{1}{5}X_{t-1} - \frac{3}{8}X_{t-2} + Z_t + \frac{7}{6}Z_{t-1} - \frac{1}{2}Z_{t-2}$

[9 marks]

(b) The (weakly) stationary process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = -0.4Y_{t-1} + 0.32Y_{t-2} + Z_t.$$

(i) Write this process in the form

$$Y_t = \sum_{k=0}^{\infty} \psi_k Z_{t-k}$$

where the coefficients $\{\psi_k : k = 0, 1, 2, \dots\}$ are defined in terms of k . You should show your working clearly.

(ii) Using your answer to (i), determine the autocovariance function of $\{Y_t : t \in \mathbb{Z}\}$.

[21 marks]

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 .

(a) The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \frac{1}{3}X_{t-1} + Z_t + \frac{8}{3}Z_{t-1} - Z_{t-2}$$

(i) Determine the spectral density function of $\{X_t : t \in \mathbb{Z}\}$.

(ii) Sketch the spectral density function from (i) for frequency values, w , where $w \in [0, \pi]$.

(iii) Using the spectral density function, determine the variance of the process $\{X_t : t \in \mathbb{Z}\}$ in terms of σ_Z^2 .

[16 marks]

(b) A linear filter is applied to the process $\{X_t : t \in \mathbb{Z}\}$ from (a) where

$$W_t = \frac{1}{2}X_t - \frac{1}{2}X_{t-1}.$$

(i) Determine the spectral density function of the process $\{W_t : t \in \mathbb{Z}\}$.

(ii) Briefly describe the effect of the linear filter on the original process $\{X_t : t \in \mathbb{Z}\}$.

[14 marks]

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 .

- (a) The process $\{X_t\}$, where t is a calendar year, measures the annual number of transactions (in tens of thousands) made on a company's online shopping website and is an ARMA(1,1) process of the form

$$X_t = \mu + \phi(X_{t-1} - \mu) + Z_t + \theta Z_{t-1}.$$

Here, μ , ϕ and θ are real-valued parameters.

Data on past annual numbers of transactions are collected and a model of the above form is fitted, with parameter estimates: $\hat{\mu} = 20.1$, $\hat{\phi} = 0.6$, $\hat{\theta} = 0.4$ and $\hat{\sigma}_Z^2 = 1.1$. The following table shows annual numbers of transactions for the years 2018–2022:

Year	2018	2019	2020	2021	2022
No. of transactions (tens of thousands)	20.1	19.6	20.5	20.3	19.8

Using the above information, provide an optimal forecast for the number of transactions in 2023. Here, the term 'optimal' implies that the forecasted value is that which minimises the mean square prediction error at a given time point. You should show your working clearly.

[18 marks]

- (b) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \lambda Y_{t-3} + Z_t$$

where λ is a real-valued parameter.

- (i) Determine the set of values of λ for which this process is (weakly) stationary. Justify your answer.
- (ii) Assuming that the process $\{Y_t : t \in \mathbb{Z}\}$ is (weakly) stationary, with appropriate justification determine the partial autocorrelation function (PACF) for this process. You may use, without proof, that the autocovariance of $\{Y_t : t \in \mathbb{Z}\}$ at lag h ($h \in \mathbb{Z}$), denoted $\gamma_Y(h)$, is

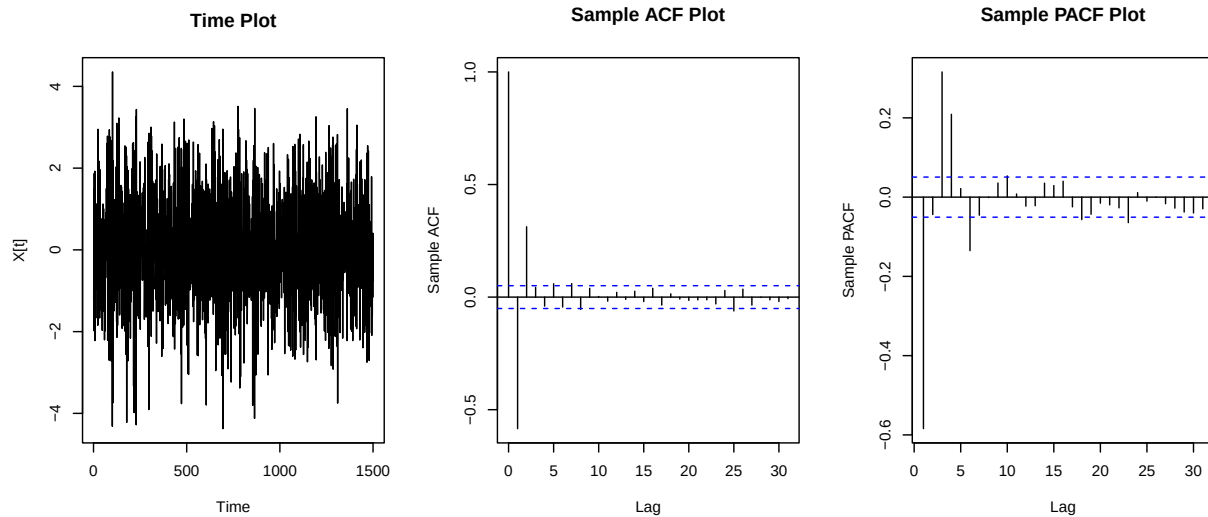
$$\gamma_Y(h) = \begin{cases} \frac{\sigma_Z^2 \lambda^{\frac{|h|}{3}}}{1 - \lambda^2} & h = 3k \text{ (where } k \in \mathbb{Z}) \\ 0 & \text{otherwise.} \end{cases}$$

- (iii) Suppose that a large sequential sample of data from this process is collected where $\lambda = 0.7$. Using these data the sample PACF is calculated for lags from 1 to 15 and a plot of the sample PACF against the lag is produced. Describe how you would expect this plot to look. Justify your answer.

[12 marks]

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_Z^2 .

- (a) A set of 1500 sequentially sampled values of the time series $\{X_t : t \in \mathbb{Z}\}$ is obtained. The figure below shows a time plot, a plot of the sample autocorrelation function (ACF) against the lag and a plot of the sample partial autocorrelation function (PACF) against the lag, for the sampled values.



- (i) Using the plots above, suggest an appropriate model for the time series. You should justify your answer and write out the equation of your model using suitable parameters but you do not need to provide parameter estimates.
- (ii) Suppose that you fit the model suggested in (i) to the sampled data. Describe how you would check that the model is a good fit to the data. Note: you do not need to refer to R commands or other computer code to answer this question.

[14 marks]

- (b) The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = Z_t + \psi Z_{t-2}$$

where ψ is a real-valued parameter such that $|\psi| < 1$.

- (i) Determine the autocovariance generating function for this process. You should show your working clearly.
- (ii) Using your answer to (i), write down the autocovariance function for the process $\{Y_t : t \in \mathbb{Z}\}$.
- (iii) The process $\{W_t : t \in \mathbb{Z}\}$ is defined

$$W_t = \frac{1}{2} (Y_t + Y_{t-1}).$$

Determine the autocovariance function for this process.

[16 marks]